Technical Datasheet - GridStore-D3

VBF-T2R1-DSH-01 | t2_r1 grid energy storage | 2026-08-25 | values mechanically sourced from parameter set / simulation outputs

Technical datasheet

Field	Value	Source
Rated capacity (Ah)	5.0 nominal (parameter set); 5.066 simulated 1C discharge	Chen2020 dump; cell/r6_d3_1c.json:capacity_ah
Nominal voltage / voltage window (V)	2.5 - 4.2 V; plateau proxy 3.722 V at fixed t = 1800 s	Chen2020 dump; r6_d3_1c.json (calc-energy 'midpoint' is a step-density artifact - design_spec section 6)
Rated energy (Wh)	18.454 (time integration of V*I at 1C)	cell/r6_d3_energy.json:energy_wh
Energy density	465.62 Wh/kg (contract caliber, electrolyte excluded); 894.87 Wh/L	cell/r6_d3_energy.json:energy_density_wh_kg / _wh_l
Maximum continuous discharge rate	1C (5.066 A) simulated; higher rates not in task contract	cell/r6_d3_1c.json
Fast-charge capability	4C charge @ 45C: plating-free (anode pot min +0.0498 V), acceptance 0.838 Ah, T_max 342.39 K (+24.2 K)	cell/r6_d3_4c.json
Operating temperature range	Verified -20C (253.15 K) to +45C (318.15 K) per simulation conditions; -20C retention 99.571% (protocol and true-soak, cell temp 266.7 K)	cell/r6_d3_lowT_ret.json; r6_d3_lowT_soak.json
Cycle life	Aging protocol (1C CC 2.5-4.2 V, SEI model): SEI 9.087 nm @100 cyc, 25.317 nm @500 cyc (limits 500/550 nm); trajectory flat. Annotation: reported per-cycle capacity variable is a PyBaMM artifact (freezes at first-discharge value); probe confirms real steps are full 1C swings (5.08 Ah dis / 4.75 Ah chg)	cell/r6_d3_aging100.json / r6_d3_aging500.json; bridge/probe_aging.py
Safety determination	4C plating-free; T_max 342.39 K during 4C charge (no contract red line); nail/overcharge/crush not simulated - N/A (beyond pure simulation boundary)	cell/r6_d3_4c.json
Dimensions and mass	1580 x 65 x 0.2008 mm; 39.63 g contract caliber (electrolyte excluded); shell Not provided	Chen2020 dump; r6_d3_energy.json